Circuit Breakers

Lecture 4

Circuit breaker

- **A circuit breaker (CB) is an automatically switchable device to protect an electrical circuit from overload or short circuit.**
- The main advantage is that operations can be reset.
- It comes in varying sizes and has a current rating.
- Circuit breakers can be categorized by voltage level (low, medium, or high) or by the method used to extinguish the arc, such as air, oil, vacuum, or SF6 gas.

A Circuit Breaker definition

- A circuit breaker is a mechanical switch that automatically operates to protect a circuit from the damage caused by fault current. It automatically breaks the circuit upon sensing a huge draw of current flow due to overloading or short circuit. It can also manually break open the circuit for maintenance or fault clearance. It can safely close and open a circuit to protect it from damage
- A circuit breaker breaks the supply to the circuit when the current exceeds its rated current. The current may exceed due to numerous reasons such as overloading, short circuit, voltage spikes, etc. Overloading occurs when the load draws a very huge current more than the rated current. A short circuit occurs when two exposed wires come into contact with each other by any means.

The Circuit Breaker function

The main function of a circuit breaker is to safely interrupt the flow of current in a circuit.

- It should momentarily withstand the fault current.
- It should safely break the circuit.
- It should quickly extinguish the arc.
- Its terminals should withstand the voltage after breaking.
- It should prevent the arc from re-striking.

The circuit breaker withstands the fault current momentarily and allows other circuit breakers to resolve the fault. The CB is designed to tolerate a specific range of fault current without damaging its terminals.

Once it detects the fault current, it trips and interrupts the current flow. It breaks open the circuit using some sort of stored mechanical energy such as a spring or a blast of compressed air, to separate the contacts. it can also use the fault current to break open the contacts using thermal expansion or an electromagnetic field using a solenoid.

The next step that comes after the separation of contacts is the arc extinction. The arc is generated between the contacts due to the high voltage between them. It can damage the CB contacts or terminals due to excessive heat generated because of high current.

Media Used for Arc Extinction

- The electrical arc tries to make the circuit, so the current still flows in it. It must be extinguished and different kinds of circuit breakers use various insulating or dielectric arc extinction mediums such as.
- Air
- Vacuum
- Insulating Oil
- Insulating gas such as SF6 (Sulphur hexafluoride)
- Other than the media being used for arc quenching, various arc extinction techniques are used to quickly and safely eliminate an arc.

Methods Used for Arc Extinction

- **Cooling of Arc**: The arc heats up the air molecule which ionizes and reduces the resistance of the air. Cooling the arc will recombine the ionized particle into its natural state and increase the dielectric strength of the air molecule. As the resistance of the medium increases, the voltage required to maintain the arc also increases and the current starts to drop resulting in arc quenching.
- **Air Blasting**: Such a method is used in the air blast circuit breaker, where the arc is quenched using a blast of compressed air. The ionized air particles are replaced with non-ionized air molecules that have higher dielectric strength. It increases the resistance thus reducing the current which leads to the extinction of the arc.

Increasing the length of the arc: The arc length is directly proportional to its voltage. Increasing the length of the arc by separating the contact terminals further apart will increase the voltage required to maintain it. Thus, it will extinguish.

Reducing the cross-section of the arc: Another technique is to reduce the cross-section of the arc by reducing the contact sizes. Therefore, the voltage required for the arc increases and extinguishes it.

Deflecting the arc: In this technique, a magnetic field is created to deflect the arc. it blows out the arc into a section of the circuit breaker called arc chute where it is cooled off and extinguished.

Dividing or splitting the arc: In this technique, the arc is split into multiple arcs by proving multiple contacts in between. The arc is split into numerous small arcs in series, which increases its length and resistance. Therefore, reducing the arc current and eventually extinguishes it.

- **Zero current quenching:** This is the most common method used in the AC circuit breaker. There are inherently multiple zero currents in an AC waveform. The circuit is opened at the exact point of zero current. So that the current does not rise to generate an arc.

- **Using a charged capacitor in parallel**: This technique is used in DC circuit breaker. The DC does not have natural zero currents. Therefore, a charged capacitor with an inductor is used in parallel to introduce an artificial zero current in the line to extinguish the arc.

The features of circuit breakers

- A circuit breaker is necessary to install on every line to protect it from any kind of hazard or disaster. Circuit breakers are manufactured by various features such as;
- Intended Voltage Applications
- Alternating or Direct Current
- Location of the installation
- Design Characteristics
- Method and medium used for current interruption (Arc Extinction)

Circuit breaker classification

- **1- Based on voltage supply type:**

➢ AC Circuit Breaker

➢ DC Circuit Breaker

- **2- Based on Voltage Level:**

➢ High voltage circuit breakers

➢ Medium-Voltage Circuit Breakers

➢ Low-Voltage Circuit Breakers

- **3- By Installation Location**

➢ Indoor Circuit Breakers

➢ Outdoor Circuit Breakers

4- Based on External Design

- Dead Tank Circuit Breakers
- Live Tank Circuit Breakers

5- According to the Operating Mechanism

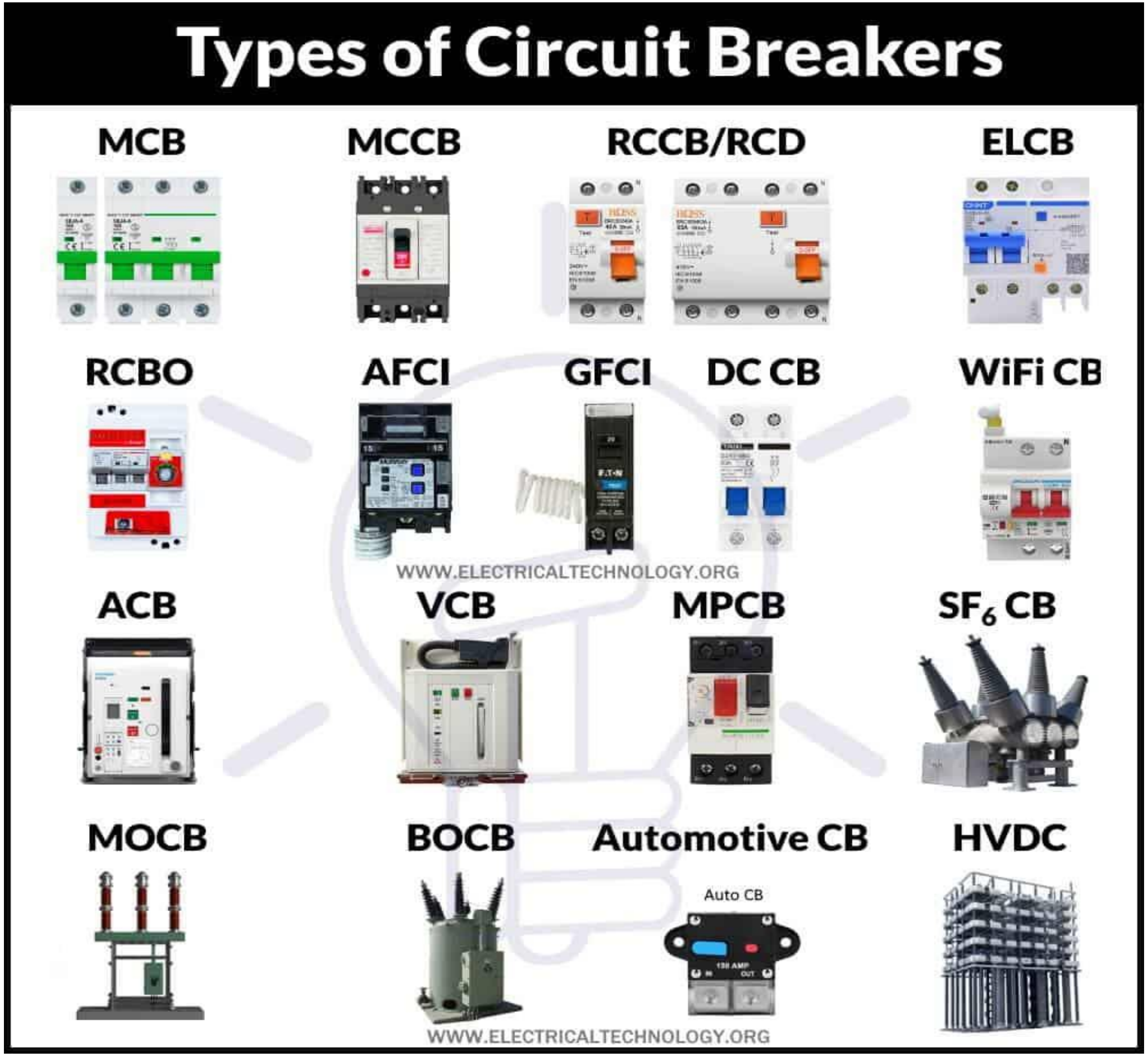
- Spring-operated Circuit Breaker
- Pneumatic-operated Circuit Breaker
- Hydraulic-operated Circuit Breaker
- Magnetic-operated Circuit Breaker
- Magnetic-hydraulic Circuit Breaker
- Air Circuit Breakers (ACB).
- Oil Circuit Breakers (OCB)
- Vacuum Circuit Breakers (VCB)
- SF6 Circuit Breakers
- Hybrid Circuit Breakers

6- Based on Structure

- Single-Pole Circuit Breakers
- Double-Pole Circuit Breakers
- Triple-Pole Circuit Breakers
- Four-Pole Circuit Breakers

7- Other Circuit Breakers

- Motor Protection Circuit Breaker (MPCB)
- Automotive Circuit Breakers
- Residual Current Device (RCD) or Residual Current Circuit Breaker(RCCB)
- Residual Current Breaker with Over-current protection (RCBO).
- Earth leakage circuit breaker (ELCB).



Types of Circuit Breakers
MCB
MCCB
RCCB/RCD
ELCB
RCBO
AFCI
GFCI
DC CB
WiFi CB
WWW.ELECTRICALTECHNOLOGY.ORG
ACB
VCB
MPCB
SF6 CB
MOCB
BOCB
Automotive CB
HVDC
Auto CB
WWW.ELECTRICALTECHNOLOGY.ORG

Circuit breaker classification in details

- There are various types of circuit breakers that are differentiated based on various characteristics.
- **1- Based on voltage supply type:**
- Circuit breakers are mainly classified into two types;

- AC Circuit Breaker
- DC Circuit Breaker

AC Circuit Breaker

- AC refers to alternating current whose voltage and current fluctuates along the zero value many times in a second. The energy at this zero point is null which can be utilized to break the circuit without generating the arc.
- Circuit breakers used for AC are quite different than in DC. The inherent zero crossings in AC provide multiple chances in a second for the arc to extinguish itself.
- The strength of the arc is directly proportional to the level of the voltage. Therefore, low voltage arcs can be easily quenched but high voltages arc require a more sophisticated approach to extinguish it. therefore, the CB are classified based on their voltage level.

DC Circuit Breaker

- As the name suggests, such CBs are used in DC circuits. DC or direct current is unidirectional, having no natural zero current. it is continuously stable, having some positive value.
- Unlike AC, which has multiple zero currents (having zero energy), that is used to quench the arc more easily.
- DC does not have such a natural zero current, so it is much harder to quench the arc.
- DC circuit breaker uses a magnet that pulls the arc to lengthen it and make it easier to quench. However, there are some low voltage circuit breakers that can be used for both AC and DC circuits.